

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2021

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2021 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. Acid + carbonate produces [1]

- A) salt + hydrogen
- B) salt + water + carbon dioxide
- C) salt + oxygen
- D) water only

2. For main-group elements, the group number indicates the [1]

- A) mass number
- B) number of electron shells
- C) number of outer-shell electrons
- D) number of neutrons

3. Convection occurs in [1]

- A) solids only
- B) fluids (liquids and gases)
- C) a vacuum
- D) crystals only

4. A magnetic material can be magnetised by [1]

- A) heating it in water only
- B) painting it red
- C) stroking with a magnet or placing it in a coil with d.c.
- D) dissolving it

5. Aerobic respiration takes place mainly in the [1]

- A) cell wall
- B) chloroplast
- C) nucleus
- D) mitochondrion

6. Ribosomes are the site of [1]

- A) photosynthesis
- B) lipid storage
- C) protein synthesis
- D) osmosis

- 7.** In humans, fertilisation normally occurs in the **[1]**
A) oviduct / Fallopian tube
B) cervix
C) ovary
D) uterus wall only
- 8.** A learner runs 20 m in 5 s. Average speed is **[1]**
A) 4 m/s
B) 100 m/s
C) 5 m/s
D) 25 m/s
- 9.** In double circulation, blood passes through the heart **[1]**
A) once per circuit of the body
B) twice per circuit of the body
C) not at all
D) only at night
- 10.** The SI unit of mass is the **[1]**
A) kilogram
B) degree Celsius
C) metre per second
D) newton
- 11.** Tuberculosis (TB) is caused by a bacterium and is spread mainly by **[1]**
A) contaminated rivers only
B) ticks
C) droplet infection (coughing, sneezing)
D) mosquitoes
- 12.** In water treatment, filtration **[1]**
A) adds fluoride only
B) kills all viruses only
C) boils the water
D) removes solid particles
- 13.** Which energy conversion is correct? **[1]**
A) chemical → kinetic in a battery at rest
B) gravitational potential → kinetic as an object falls
C) sound → nuclear always
D) light → mass always
- 14.** A human sperm cell has a flagellum so that it can **[1]**
A) carry oxygen
B) store yolk
C) swim towards the egg
D) photosynthesise
- 15.** The SI unit of force is the **[1]**
A) watt
B) kilogram
C) newton
D) joule

16. Blue litmus paper in an acid [1]

- A) turns green
- B) stays blue
- C) turns red
- D) bleaches white always

17. The law of reflection states that [1]

- A) angle of incidence = angle of reflection
- B) a mirror absorbs all light
- C) angle of incidence = angle of refraction always
- D) light always bends towards the normal in air

18. A gas has [1]

- A) particles that cannot move
- B) fixed volume, no fixed shape
- C) fixed shape and volume
- D) no fixed shape or volume

19. In a liquid, particles [1]

- A) do not move
- B) are close but can slide past one another
- C) are fixed in a lattice
- D) are far apart with no forces

20. Anaerobic respiration in yeast produces [1]

- A) ethanol and carbon dioxide
- B) starch
- C) lactic acid only
- D) oxygen

21. The arrows in a food chain show [1]

- A) wind direction
- B) the direction the animal walks
- C) the age of the organism
- D) the direction of energy flow

22. Which is true of magnets? [1]

- A) Like poles attract
- B) North-south attract
- C) North-north attract
- D) South-south attract

23. The gravitational force on an object is its [1]

- A) mass
- B) density
- C) volume
- D) weight

24. Chlorine is added to drinking water to [1]

- A) kill microorganisms
- B) add hardness
- C) make it cloudy
- D) remove all ions

25. Alkalis in aqueous solution contain [1]

- A) Cl^- only
- B) OH^-
- C) Na^+ only
- D) H^+

26. Water enters a root hair cell mainly by [1]

- A) photosynthesis
- B) ejection
- C) transpiration pull only
- D) osmosis

27. The rate of a reaction increases when [1]

- A) concentration decreases
- B) the surface area of a solid reactant increases
- C) temperature decreases
- D) the catalyst is removed

28. Sweat contains [1]

- A) haemoglobin
- B) starch
- C) only pure oil
- D) water, salts and a little urea

29. A plant cell in a concentrated sugar solution may become [1]

- A) a gamete
- B) plasmolysed
- C) photosynthetic
- D) turgid

30. Electrolysis of aqueous copper(II) sulfate with copper electrodes is used to [1]

- A) extract aluminium
- B) make sodium
- C) liquefy air
- D) purify copper

31. A neutron has [1]

- A) charge -1
- B) no charge
- C) charge $+1$
- D) charge $+2$

32. A learner runs 20 m in 4 s. Average speed is [1]

- A) 80 m/s
- B) 4 m/s
- C) 5 m/s
- D) 24 m/s

33. A high temperature denatures an enzyme because it [1]

- A) turns it into starch
- B) changes the shape of the active site
- C) increases pH only
- D) adds a coenzyme

34. Sodium chloride is held together by **[1]**

- A) covalent bonds only
- B) hydrogen bonds only
- C) ionic bonds
- D) metallic bonds only

35. Potassium is stored under oil because it **[1]**

- A) dissolves oil
- B) is radioactive
- C) reacts vigorously with air and water
- D) is a gas

36. Kinetic energy depends on **[1]**

- A) mass and speed
- B) colour only
- C) volume only
- D) charge only

37. An object has mass 50 g and volume 20 cm³. Its density is **[1]**

- A) 70 g/cm³
- B) 20 g/cm³
- C) 1000 g/cm³
- D) 2 g/cm³

38. Phloem tissue transports **[1]**

- A) sucrose and amino acids
- B) water only upwards
- C) oxygen
- D) soil particles

39. Metals are good thermal conductors because they have **[1]**

- A) a vacuum inside
- B) only covalent bonds
- C) no particles
- D) free electrons that transfer energy

40. A leaf that has photosynthesised is tested with iodine. A blue-black colour shows **[1]**

- A) starch
- B) protein
- C) lipid
- D) reducing sugar

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).